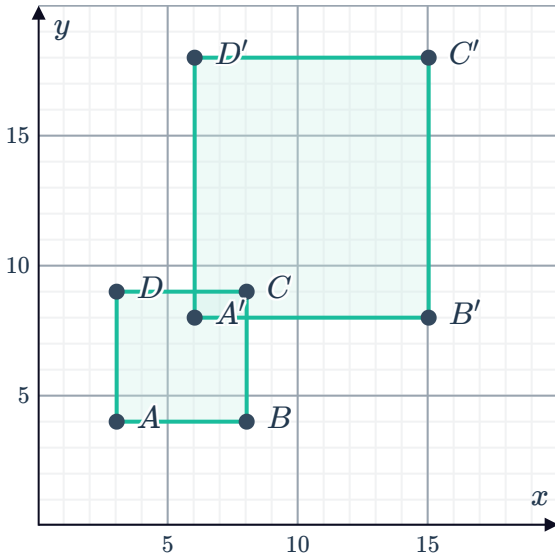


Dilations

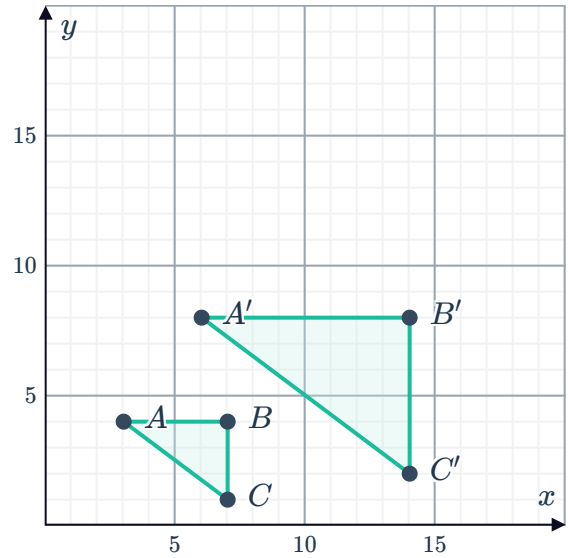


- 1 Determine whether the following graphs show a dilation. If yes, find the scale factor.

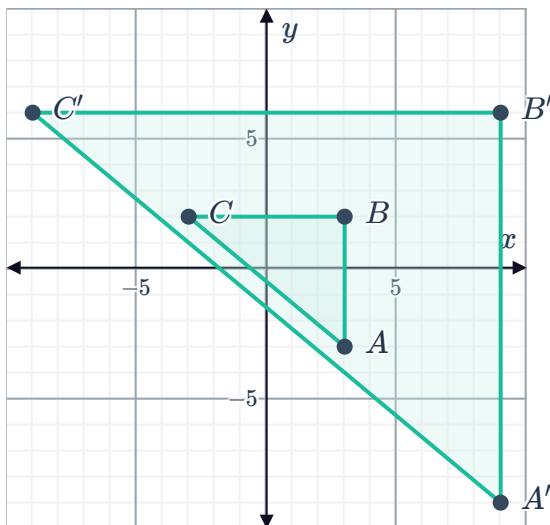
a



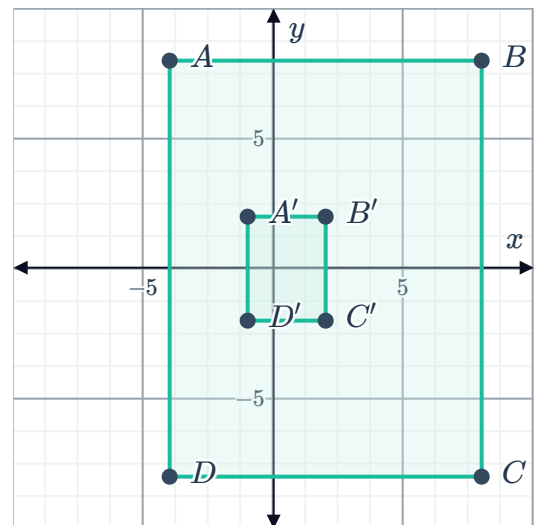
b



c



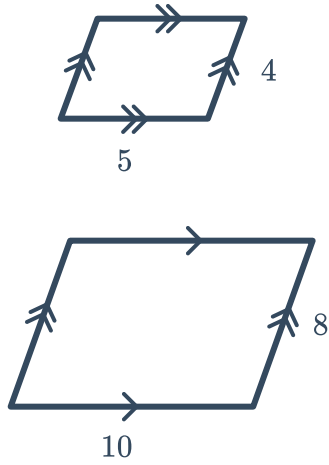
d



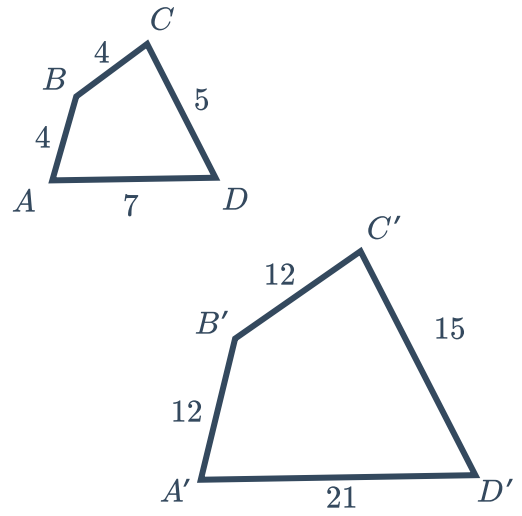
2 Find the scale factor for each pair of similar figures:



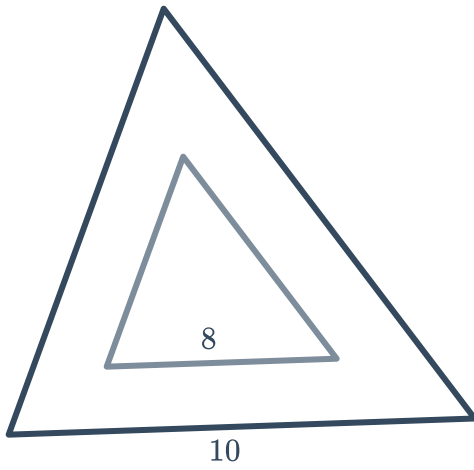
a



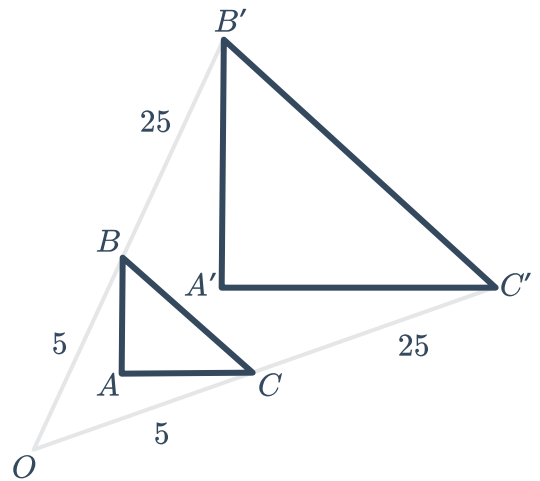
b



c



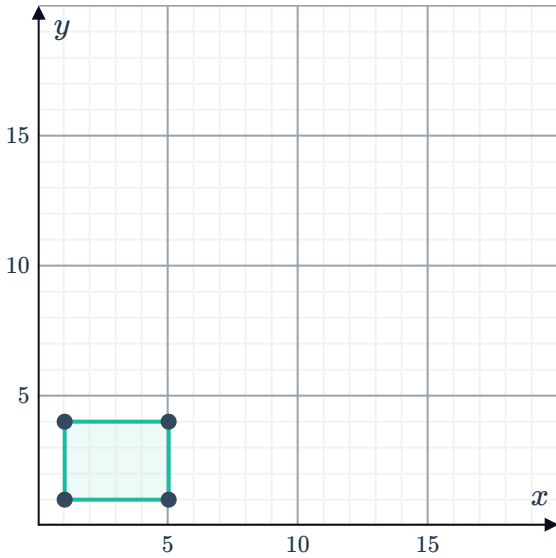
d



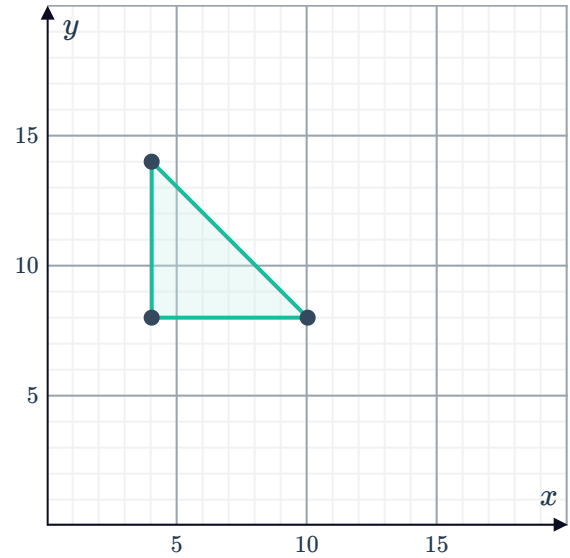
3 Dilate the figure by the given factor using the origin as the center of dilation:



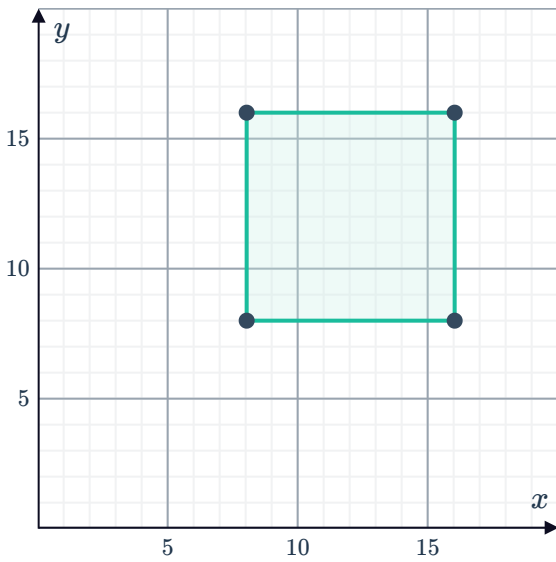
a Scale factor of 3



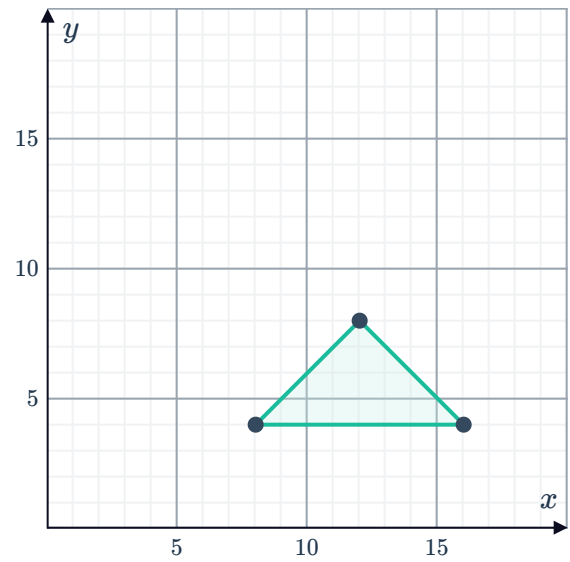
b Scale factor of 4



c Scale factor of $\frac{1}{2}$



d Scale factor of $\frac{5}{4}$

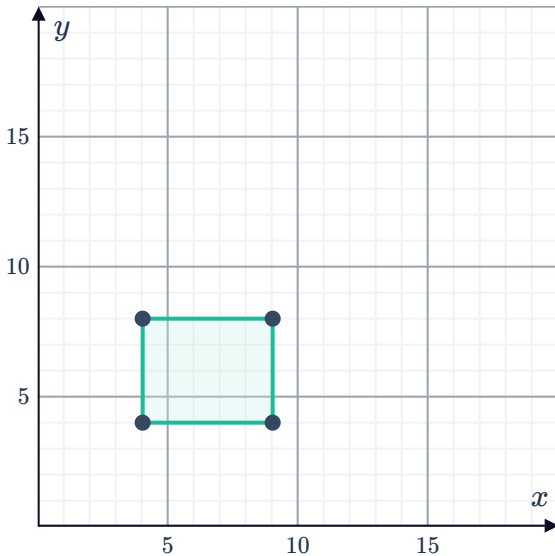


4 Dilate each figure using the given scale factor and center of dilation.



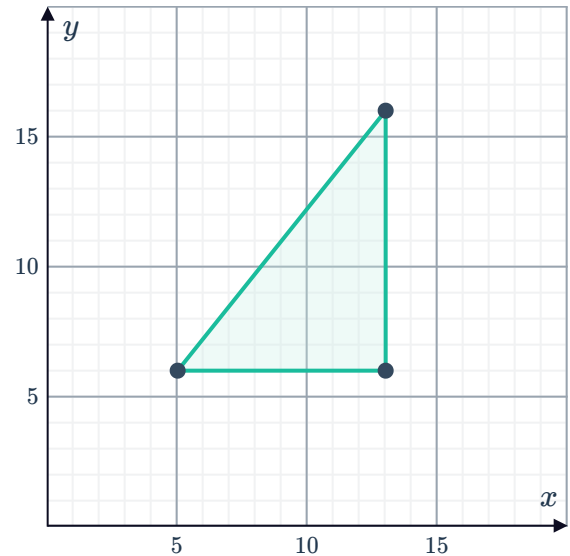
a Scale factor: 2

Center of dilation: $A(2, 3)$



b Scale factor: $\frac{1}{2}$

Center of dilation: $A(3, 2)$



5 A triangle with vertices $A(-7, -5)$, $B(3, -5)$ and $C(-5, 9)$ is dilated by a factor of 0.5 using the origin as the center of dilation. What are the coordinates of the vertices of the dilated triangle?

6 A rectangle with vertices $P(-5, 3)$, $Q(3, 3)$, $R(3, -5)$ and $S(-5, -5)$ is dilated using the origin as the center of dilation. The vertices of the new rectangle are $P'(-25, 15)$, $Q'(15, 15)$, $R'(15, -25)$ and $S'(-25, -25)$.

Find the scale factor.